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Modul 12: Shifting operations

Stand: 18. Februar 2026

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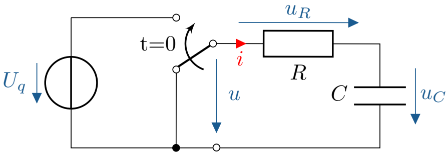
Introduction to switching operations

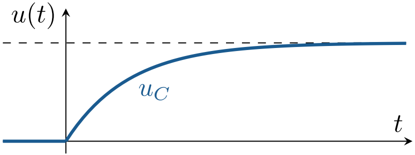
Learning objectives: Introduction

The students learn:

- Recognise switching operations as non-stationary states following switching actions
- Understand switching operations in the context of other compensation processes
- Recognise challenges and opportunities associated with switching operations
- Distinguish between the switching behaviour of ideal and real switches

Settlement procedures





[1]Image: Bartolomeo Pinelli, detail from *A Peasant Family Cooking over a Campfire*, Licence CC0 1.0
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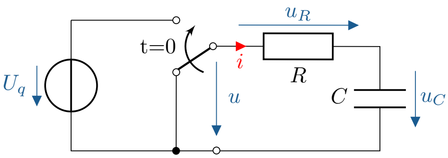
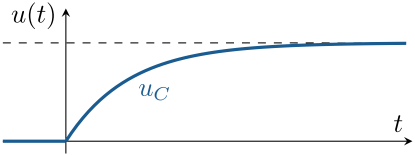
Settlement procedures





Analogy: Heating water[1]

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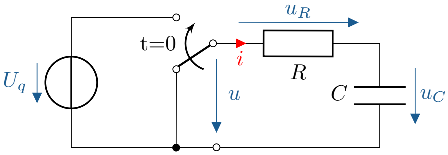

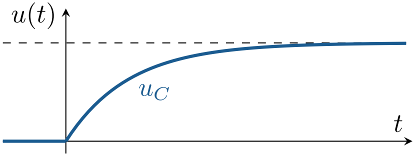

Analogy: Heating water[1]

Reasons for balancing transactions:

- Switching operations
- DC: Change in voltage/current
- AC: Change in frequency/amplitude/phase

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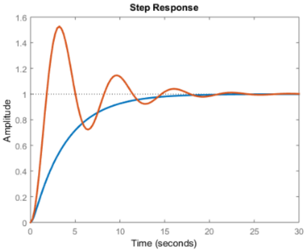
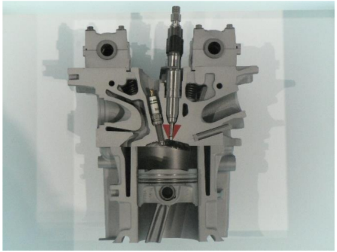
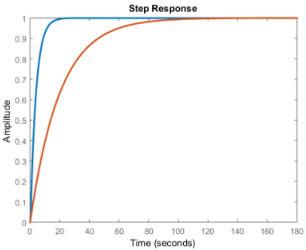
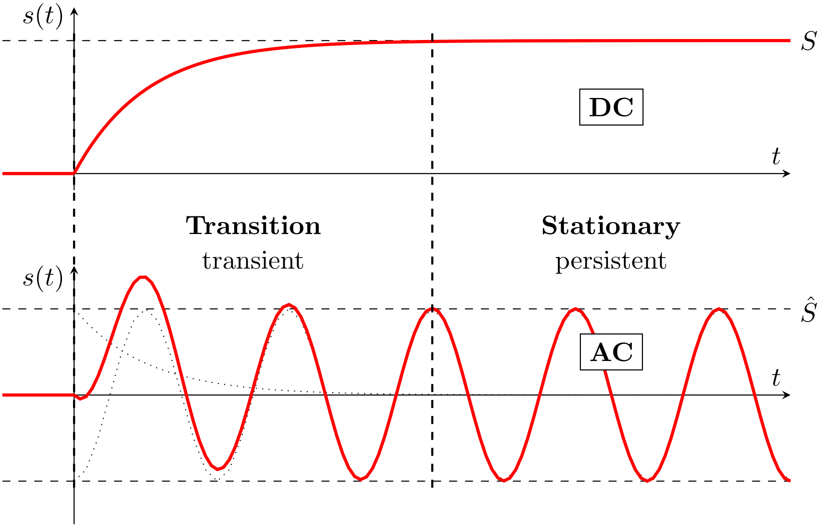
Settlement procedures - Examples



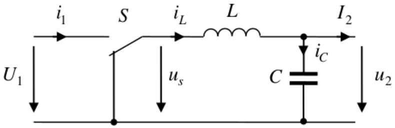




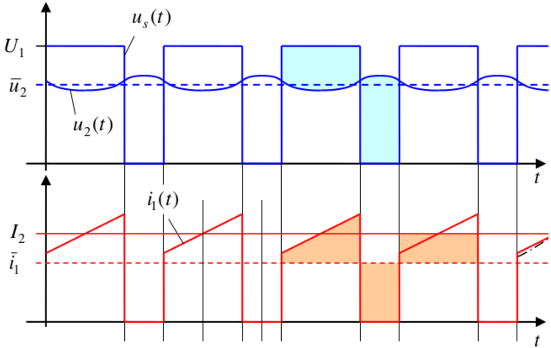
Example of switching operation with DC and AC voltage.



Example from power electronics:



Step-down converter with smoothing capacitor[2]

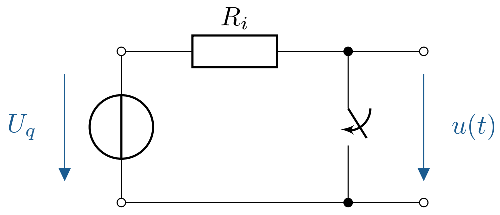


Power converters convert electrical energy into electrical energy. The conversion (e.g. voltage, current, frequency) is achieved by means of clocked switching.

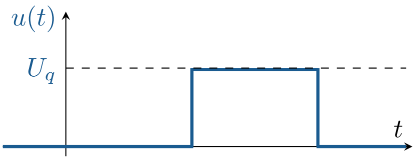
Other examples:

Switching on/off, AD/DA converter, bicycle light with capacitor, ...

[2]Schematic diagram and time sequence: Joachim Böcker, GET2, University of Paderborn, modified (abridged)

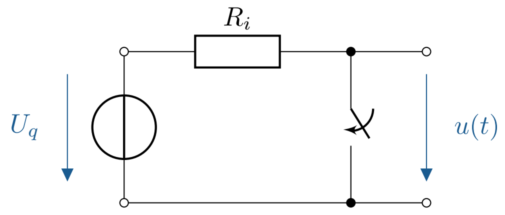


Switch voltage $u(t)$



Ideal: lossless, instantaneous

Comparison of ideal and real switches



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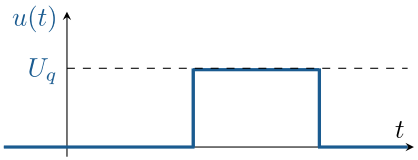

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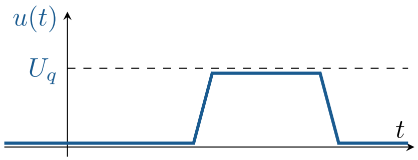
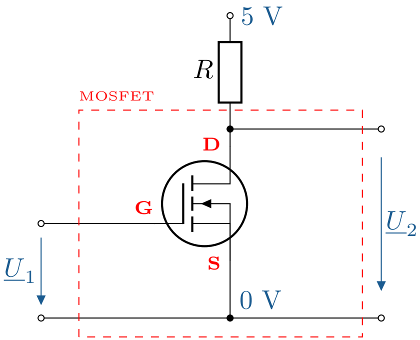


Real: Losses, latencies



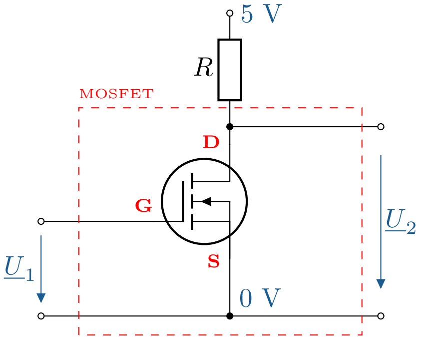
no parasitic effects

Ideal:

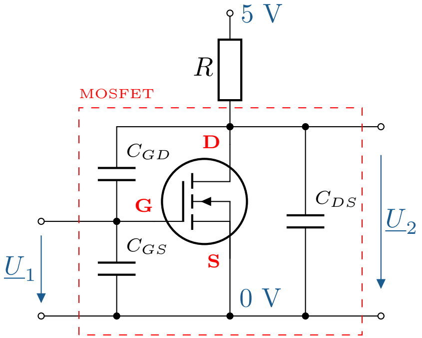
Here: Settlement procedures <u>after</u> Switching (switching operations) in focus

Real: Settlement procedures <u>during</u> switching

Confine.: Assumption of ideal switches



Ideal: no parasitic effects



Real: with parasitic capacities

Here:	Settlement procedures	after	Switching (switching operations) in focus
Real:	Settlement procedures	during	switching
Confine.:	Assumption of ideal switches		

Fundamentals of calculating switching operations

Learning objectives: Fundamentals

The students learn:

- Know and understand fundamental concepts (definitions)
- Be familiar with calculation methods in the time domain and frequency domain (Laplace)
- Set up differential equations for circuit quantities of LTI circuits
- Derive properties of differential equations from circuit properties
- Determine time responses during switching processes (solve differential equations)

Definitions

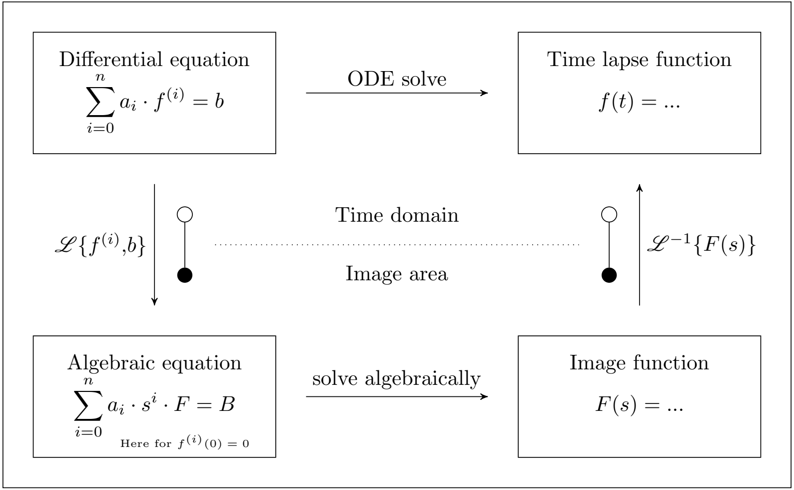
G.	Terms	Definitions
General.	Switch	„set to an (operating) state by pressing a switch“[‡]
	Switch	Component, in this case for opening or closing an electrical connection
Properties	Periodic[1]	„at regular intervals, regularly [occurring, recurring]“[‡]
	Time-invariant[1]	invariant/unchanging over time
	Usually[2]	Derivative(s) according to an independent variable (e.g. time t), cf. partial
	Homogeneous[2]	Usually in the form: $\sum a_i(t) \frac{\mathrm{d}^i}{\mathrm{d}t^i} f_i(y(t)) \stackrel{!}{=} 0$ with interference function $b = 0$
	Linear[2]	Usually in the form: $\sum a_i(t) \frac{\mathrm{d}^i}{\mathrm{d}t^i} y(t) = b(t)$ d.h. mit $\frac{\mathrm{d}^i}{\mathrm{d}t^i} y(t)$ linear
	Order[2]	n $[1]$ i.e. at most derivation of order n in DGL
Procedures	Compensation process	Process in a system that strives for a steady state
	Settling-in period	Compensation process, oscillating, endergon (not spontaneous)
	Switching process	non-steady state after switching
	Transient	Compensation process (synonym), cf. transient (adj.)

[1]generally [2]relating to DGL

‡Duden online

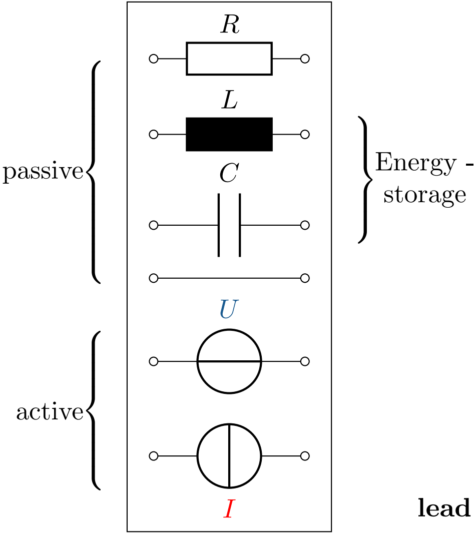
Calculation methods for switching operations

Switching operations can be calculated in the time and image domain.



Both methods are based on solving differential equations.

Circuit



Equations

Components

Resistance $u_R = R \cdot i_R$

Inductance $u_L = L \cdot \frac{\mathrm{d}}{\mathrm{d}t} \cdot i_L$

Capacity $i_C = C \cdot \frac{\mathrm{d}}{\mathrm{d}t} \cdot u_C$

Energy storage

Kirchhoff

Knot: $\sum i = 0$

Mesh: $\sum u = 0$

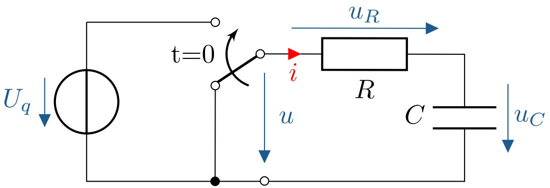
Energy storage in circuits lead to switching operations and DGLs.

Differential equations

Possible approaches to setting up differential equations:

Knot/mesh equation. $\sum_{j=1}^{n} i_j = 0, \quad \sum_{k=1}^{m} u_k = 0$	mit *	Component Gl. resolve to target size ggf. $\frac{\mathrm{d}}{\mathrm{d}t}$, $\int \mathrm{d}t$ apply	$\longrightarrow$	Differential equation $\sum_{i=0}^{n} a_i \cdot f^{(i)} = b$
Complex alternating current-calculation $\downarrow$		Time domain Frequency range		$\uparrow$ $(\mathrm{j}\omega)^i \mathrel{\widehat{=}} \left(\frac{\mathrm{d}}{\mathrm{d}t}\right)^i$
Knot/mesh equation. $\sum_{j=1}^{n} \underline{I}_j = 0, \quad \sum_{k=1}^{m} \underline{U}_k = 0$	mit *	Impedance equation. resolve to target size Power/voltage divider poss.	$\longrightarrow$	Algebraic equation $\sum_{i=0}^{n} a_i \cdot (\mathrm{j}\omega)^i \cdot \underline{F} = B$

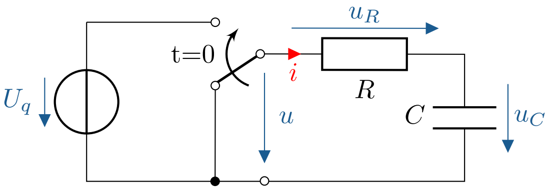
*knoten-/maschenweise vorgehen



Time domain: Gesucht sei $u_C(t \geq 0)$.

$t \geq 0$	**Kirchhoff:**	**Components:**
Knot:	$i_R = i_C = i$	$i_C = C \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C$
Mesh:	$u_R + u_C = u$	$u_R = R \cdot i_R$

Differential equationsset up, example



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$$R \cdot i + u_C = U_q$$

ODE: $RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C + u_C = U_q$ 1. Ord.

Differential equationsset up, example

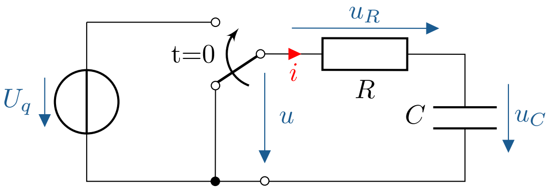


Image area: Wanted $u_C(t \geq 0)$.

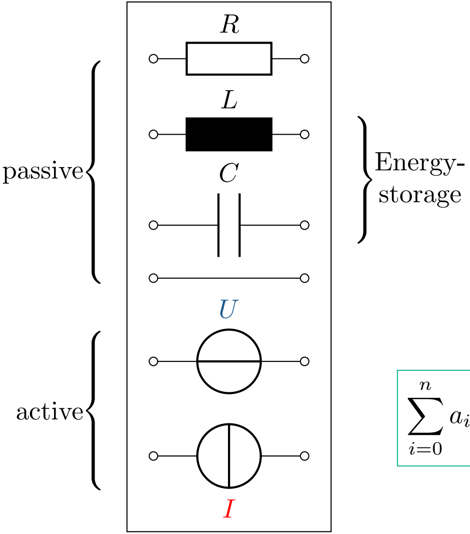
$$\underline{U}_C = \underline{U}_q \cdot \frac{\frac{1}{\mathrm{j}\omega C}}{R + \frac{1}{\mathrm{j}\omega C}}$$

$$\left(R + \frac{1}{\mathrm{j}\omega C}\right) \cdot \underline{U}_C = \underline{U}_q \cdot \frac{1}{\mathrm{j}\omega C}$$

$$(\mathrm{j}\omega \cdot RC + 1) \cdot \underline{U}_C = \underline{U}_q$$

DGL : $RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C + u_C = U_q$

Circuit



Properties

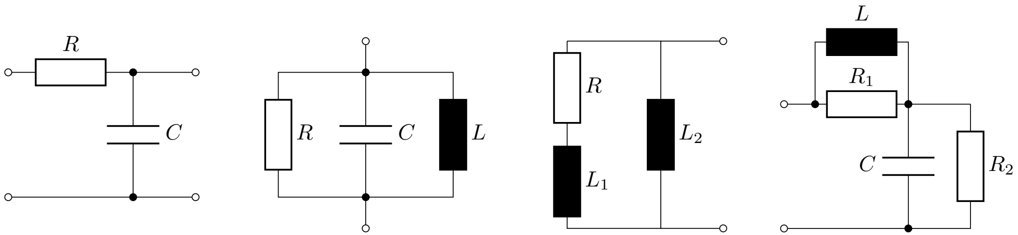
System
- Energy-storage → ODE
- linear → linear
- time-invariant → constant Coefficients

$$\boxed{\sum_{i=0}^{n} a_i \cdot \frac{\mathrm{d}^i}{\mathrm{d}t^i}\, y(t) = b(t)}$$

ODE-Sizes

n	:	Order of ODE
a_i	:	Const. coefficients
$y(t)$	:	Circuit size
$b(t)$	:	Distur. var.

Order of differential equations



1 En.sp. $\rightarrow$ 1. Ord. 2 En.sp. $\rightarrow$ 2. Ord. 2 En.sp. $\rightarrow$ 2. Ord. 2 En.sp. $\rightarrow$ 2. Ord.

Number of energy storage units§ $=$ ODE regulations

§nicht zusammenfassbar

Calculation method in the time domain

Every ODE has an infinite number of solutions.

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A unique solution for ODE n-th order requires n **Initial conditions**.

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With:

Capacity: $E_{el} = \frac{1}{2}C \cdot u^2 \implies u = steadily$ (Electric ener. storage)

Inductance: $E_{mag} = \frac{1}{2}L \cdot i^2 \implies i = steadily$ (Mag. ener. storage)

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applies immediately before and after a switching point:

$$u_{C,t-} = u_{C,t+} \qquad i_{L,t-} = i_{L,t+}$$

with $$\lim_{\pm\infty \to t} u_C(t) = u_{C,t\pm} \qquad \lim_{\pm\infty \to t} i_L(t) = i_{L,t\pm}$$

Decomposition of compensation processes and solution of inhomogeneous differential equations

Disassembly of the **System size** $y(t)$ to:

steady state $y_\mathrm{e}(t)$

and **volatile state** $y_\mathrm{f}(t)$.

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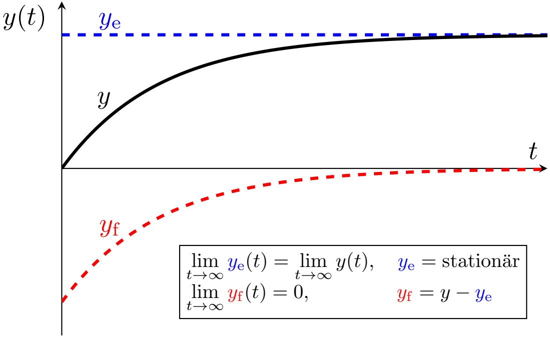

$$\lim_{t\to\infty} y_\mathrm{e}(t) = \lim_{t\to\infty} y(t), \quad y_\mathrm{e} = \text{stationär}$$
$$\lim_{t\to\infty} y_\mathrm{f}(t) = 0, \quad y_\mathrm{f} = y - y_\mathrm{e}$$

Decomposition of compensation processes and solution of inhomogeneous differential equations

Disassembly of the **System size** $y(t)$ to:
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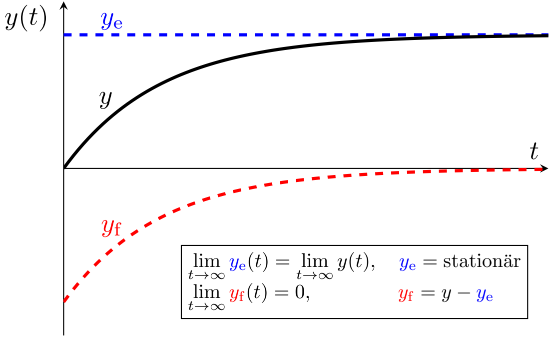
Corresponds to the breakdown of y in:
a **particular solution** y_p
and the **general homogeneous solution** y_h



$$y(t) - \underbrace{y_\mathrm{e}(t)}_{y_\mathrm{p}(t)} = \underbrace{y_\mathrm{f}\,(t)}_{y_\mathrm{h}(t)}$$

Decomposition of compensation processes and solution of inhomogeneous differential equations

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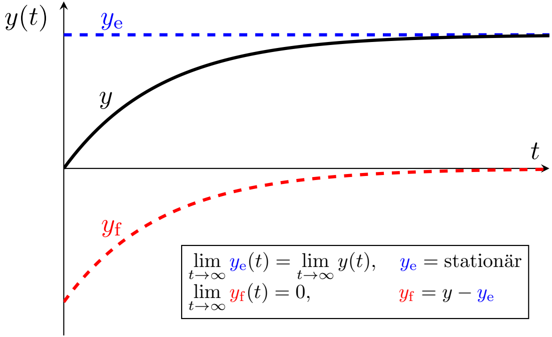

$$y(t) - \underbrace{y_\mathrm{e}(t)}_{y_\mathrm{p}(t)} = \underbrace{y_\mathrm{f}(t)}_{y_\mathrm{h}(t)}$$

with

$$\begin{aligned} &\sum a_i \cdot \frac{\mathrm{d}^i}{\mathrm{d}t^i}\, y &= b(t) \\ -\, &\sum a_i \cdot \frac{\mathrm{d}^i}{\mathrm{d}t^i}\, y_\mathrm{e} &= b(t) \\ \hline =\, &\sum a_i \cdot \frac{\mathrm{d}^i}{\mathrm{d}t^i}\, y_\mathrm{f} &= 0 \end{aligned}$$

Homogeneous ODE: $$\sum_{i=0}^{n} a_i \cdot \frac{\mathrm{d}^i}{\mathrm{d}t^i}\, y_\mathrm{h} = 0$$ (weighted, linear, const. coeff.)

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Approach: $$y_\mathrm{h}(t) = K \cdot \mathrm{e}^{\lambda t}$$ (Exponential funct.)

Char. Polynom: $$\sum_{i=0}^{n} a_i \cdot \lambda^i = 0 \quad \Rightarrow \quad \lambda_i = \ldots$$ (Eigenvalues: λ)

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1. Solution approach: $y_\mathrm{h}(t) = \sum_{i=1}^{n} K_i \cdot \mathrm{e}^{\lambda_i t}$ for $\lambda_i \neq \lambda_{i-1} \ \forall \ i$

2. Solution approach: $y_\mathrm{h}(t) = \sum_{j=1}^{m} K_j \cdot t^{j-1} \cdot \mathrm{e}^{\lambda t}$ for $\lambda_j = \lambda \ \forall \ j$

Calculation of the homogeneous solution

Homogeneous ODE: $\sum_{i=0}^{n} a_i \cdot \frac{\mathrm{d}^i}{\mathrm{d}t^i} y_\mathrm{h} = 0$ (weighted, linear, const. coeff.)

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gen. homog. solution: $y_\mathrm{h}(t) = \; \ldots \; = y_\mathrm{f}$ (Linear combination)

Example: Linearly independent solutions of a differential equation

Find the general solution of the following homogeneous fifth-order differential equation:[5]

$$\frac{\mathrm{d}^5}{\mathrm{d}t^5}y(t) + 7\frac{\mathrm{d}^4}{\mathrm{d}t^4}y(t) + 26\frac{\mathrm{d}^3}{\mathrm{d}t^3}y(t) + 62\frac{\mathrm{d}^2}{\mathrm{d}t^2}y(t) + 8\frac{\mathrm{d}}{\mathrm{d}t}y(t) + 75y(t) = 0$$

[5] Slightly modified from: Manfred Albach, *Elektrotechnik 2*, Pearson Deutschland GmbH 3. Auflage, 2020.

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The characteristic polynomial:

$$\lambda^5 + 7\cdot\lambda^4 + 26\cdot\lambda^3 + 62\cdot\lambda^2 + 85\cdot\lambda + 75 = 0$$

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has one simple and two double, conjugate, complex zeros:

$$\lambda_1 = -3, \quad \lambda_2 = \lambda_3 = -1 + 2\mathrm{j}, \quad \lambda_4 = \lambda_5 = -1 - 2\mathrm{j}$$

[5] Slightly modified from: Manfred Albach, *Elektrotechnik 2*, Pearson Deutschland GmbH 3. Auflage, 2020.

Example: Linearly independent solutions of a differential equation

Find the general solution of the following homogeneous fifth-order differential equation:[5]

$$\frac{\mathrm{d}^5}{\mathrm{d}t^5}y(t) + 7\frac{\mathrm{d}^4}{\mathrm{d}t^4}y(t) + 26\frac{\mathrm{d}^3}{\mathrm{d}t^3}y(t) + 62\frac{\mathrm{d}^2}{\mathrm{d}t^2}y(t) + 8\frac{\mathrm{d}}{\mathrm{d}t}y(t) + 75y(t) = 0$$

The characteristic polynomial:

$$\lambda^5 + 7\cdot\lambda^4 + 26\cdot\lambda^3 + 62\cdot\lambda^2 + 85\cdot\lambda + 75 = 0$$

has one simple and two double, conjugate, complex zeros:

$$\lambda_1 = -3, \quad \lambda_2 = \lambda_3 = -1 + 2\mathrm{j}, \quad \lambda_4 = \lambda_5 = -1 - 2\mathrm{j}$$

The general solution is:

$$y_{\mathrm{h}}(t) = K_1 \cdot \mathrm{e}^{-3t} + (K_2 + K_3 \cdot t) \cdot \mathrm{e}^{-t}\sin(2t) + (K_4 + K_5 \cdot t) \cdot \mathrm{e}^{-t}\cos(2t)$$

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Calculation of the particular solution

The **steady state** y_e corresponds to the **particular solution** y_p.

In LZI systems, the following applies:

DC Excitation:	$y_\mathrm{p}(t) = \text{const.}$	(Equal size)
AC-Excitation:	$y_\mathrm{p}(t) = \hat{Y} \cdot \sin(\omega t + \varphi)$	(Variable quantity)

This means that the well-known **Network calculation methods applicable**.

DC: Capacities such as open circuits and inductances such as short circuits

AC: Complex alternating current calculation (impedances, admittances)

Key point: Procedure for ODE

1. Set up differential equation (lin., ord.) $\sum a_i \cdot \frac{\mathrm{d}^i y}{\mathrm{d}t^i} = b$
2. Transient state (homog. sol.) $y_\mathrm{f} \overset{*}{=} \sum K_i' \cdot \mathrm{e}^{\lambda_i t}$
3. Steady-state condition (part. sol.) $y_\mathrm{e} = y(t \to \infty)$
4. Superposition (general sol.) $y = y_\mathrm{f} + y_\mathrm{e}$
5. Determine constant(s) (initial cond.) $i_L, u_C = \mathit{continuous}$

*with $K_i' = \begin{cases} K_i & \text{for } \lambda_i \neq \lambda_j \quad \forall \quad i,j & \text{(simple NS)} \\ K_i \cdot t^k & \text{for } \lambda_i = \lambda_{i-k} \quad \text{with} \quad k \in [0,\, 1,\, \ldots,\, m-1] & \text{(m-fold zero, k-th identical eigenvalue)} \end{cases}$

Learning objectives: Switching operations

The students learn:

- Calculate switching processes in the time domain
- Know and understand the switching behaviour of RC, RL and RLC elements
- Know the possibility of minimising switching processes with AC excitation
- Understand and analyse the oscillation behaviour of oscillating circuits

Switching operations With direct voltage

Switching operations for direct voltage are:

- Charging operations,
- Unloading operations.

Switching operations With direct voltage

Switching operations for direct voltage are:

- Charging operations,
- Unloading operations.

Calculation initially for:

- a capacity C
- an inductance L

each in series with a resistor R.

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RC or RL element $\Longrightarrow$ **ODE** 1st order

$$a_1 \cdot \frac{\mathrm{d}}{\mathrm{d}t} y(t) + a_0 \cdot y(t) = b$$

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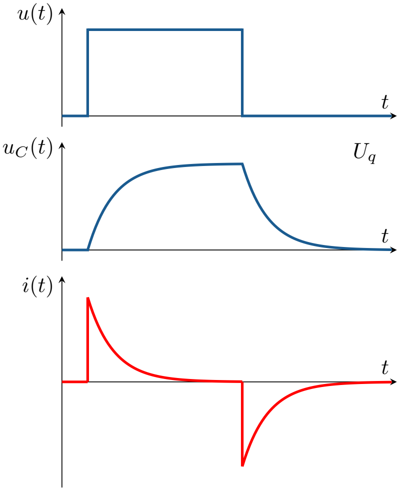
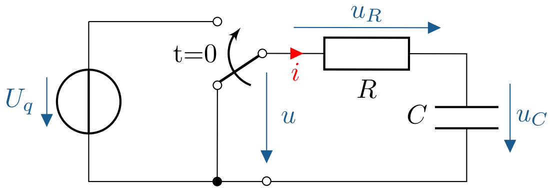
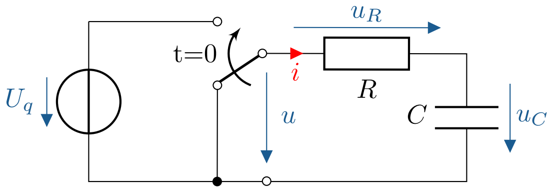


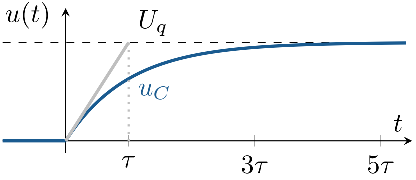
Switching operations in RC circuit (DC)



First-order ODE $\Rightarrow$ Exponential curve.

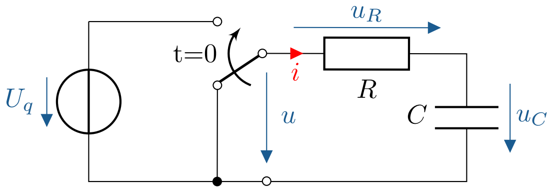
Switching operations In terms of capacity, Time constant

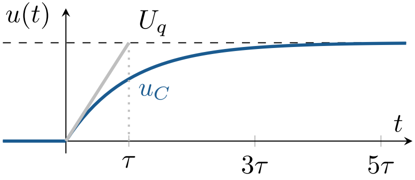




First-order ODE $\Rightarrow$ Exponential curve.

$$u_C(t) = U_q\left(1 - \mathrm{e}^{-t/\tau}\right)$$





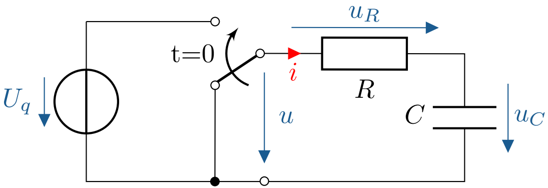
First-order ODE $\Rightarrow$ Exponential curve.

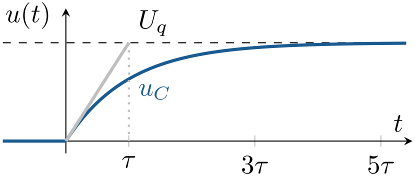
$$u_C(t) = U_q\left(1 - \mathrm{e}^{-t/\tau}\right)$$

Die **Time constant** τ, $\quad [\tau] = 1\,\mathrm{s}$:

- determines the steepness of exponential curves.
- is the time after which the tangent reaches its final value.
- is $\tau = RC$ for the RC element.

Switching operations In terms of capacity, Time constant





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- determines the steepness of exponential curves.
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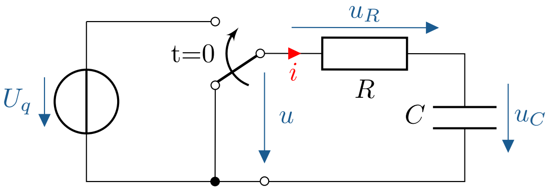
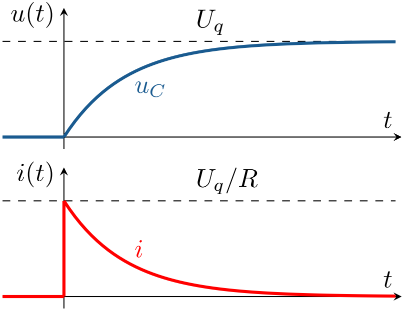
- is $\tau = RC$ for the RC element.

Typ. values Time constant τ

t	τ	$3\,\tau$	$5\,\tau$
$1 - \mathrm{e}^{-t/\tau}$	63,2%	95,0%	99,3%
$\mathrm{e}^{-t/\tau}$	36,8%	5,0%	0,7%

Rule of thumb: „Settled in“ to $5\,\tau$

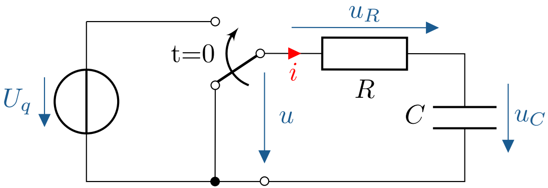




given: $u_C(0)$, U_q, R, C

wanted: $u_C(t)$

1. **ODE** $$U_q = RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C + u_C$$
2. **Homogeneous solution** $$u_{C,h} = K \cdot \mathrm{e}^{-t/\tau} \quad \text{with } \tau = RC$$
3. **Particular solution** $$u_{C,p} = U_q$$
4. **Non-homog. solution** $$u_C = K \cdot \mathrm{e}^{-t/\tau} + U_q$$
5. **Determine constants** $$u_C = U_q \left(1 - \mathrm{e}^{-t/\tau}\right)$$

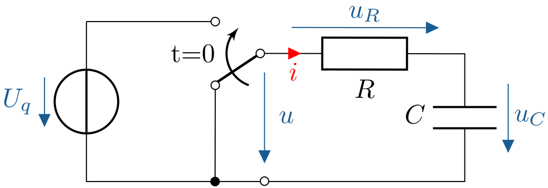


First-order ODE:

$$a_1 \cdot \frac{\mathrm{d}}{\mathrm{d}t} y(t) + a_0 \cdot y(t) = b$$

1.) Set up ODE: for $u_C(t)$ $(t \geq 0)$

Based on the mesh equation, u_R is expressed as a function of u_C:



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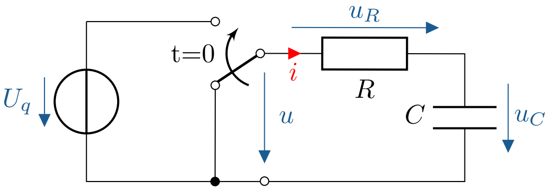

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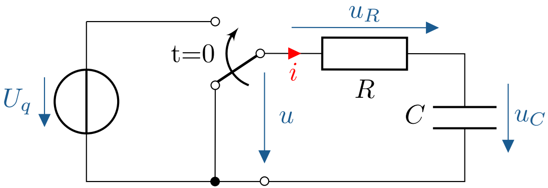
$R \cdot i + u_C = U_q$ with $u_R = R \cdot i$

Charging process In terms of capacity



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$R \cdot i + u_C = U_q$ with $u_R = R \cdot i$

ODE : $R \cdot C \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C + u_C = U_q$ with $i = C \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C$

2.) Homogeneous solution, gen.: (fleeting) (without interference)

ODE$_h$: $$RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_{C,h} + u_{C,h} = 0$$ Approach: $u_{C,h} = K \cdot \mathrm{e}^{\lambda t}$

$$K \cdot (RC \cdot \lambda \mathrm{e}^{\lambda t} + \mathrm{e}^{\lambda t}) = 0$$

char. Polynom: $$\Rightarrow \quad RC \cdot \lambda + 1 = 0 \qquad \Rightarrow \quad \lambda = -\frac{1}{RC}$$

hom. Sol. : $$u_{C,h} = K \cdot \mathrm{e}^{-t/\tau}$$ with $\tau = RC$

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hom. Sol. : $$u_{C,h} = K \cdot \mathrm{e}^{-t/\tau}$$ with $\tau = RC$

3. Particular solution: (settled in) (for $t \to \infty$)

ODE$_p$: $$\cancel{RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_{C,p}} + u_{C,p} = U_q$$ with $\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t \to \infty} = 0$

part. Sol. : $$u_{C,p} = U_q$$

4.) Non-homogenous solution, gen.: (Superimposition)

$$u_C(t) = u_{C,h}(t) + u_{C,p}(t) \qquad = u_{C,f}(t) + u_{C,e}(t)$$

gen. sol.: $$u_C(t) = K \cdot \mathrm{e}^{-t/\tau} + U_q$$

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$$u_C(t) = u_{C,h}(t) + u_{C,p}(t) \qquad = u_{C,f}(t) + u_{C,e}(t)$$

gen. sol.: $$u_C(t) = K \cdot \mathrm{e}^{-t/\tau} + U_q$$

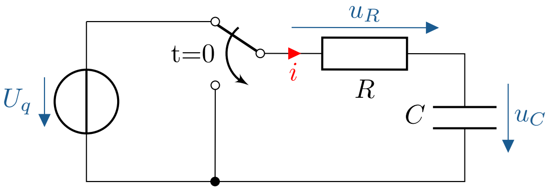
5. Determine constants: (Set initial condition)

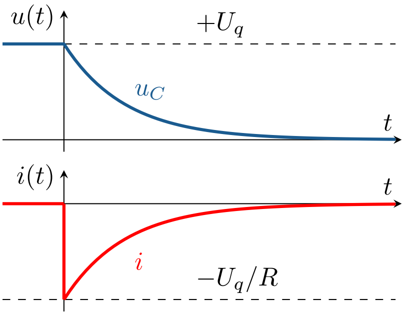
Initial cond. $$u_C(0) = K + U_q \stackrel{!}{=} 0 \qquad \Rightarrow \qquad K = -U_q$$

$$u_C(t) = -U_q \cdot \mathrm{e}^{-t/\tau} + U_q$$

special gen. sol.: $$u_C(t) = U_q \left(1 - \mathrm{e}^{-t/\tau}\right)$$

Current: $$i_C(t) = \frac{U_q}{R} \cdot \mathrm{e}^{-t/\tau}$$ with $$i_C = C \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C$$

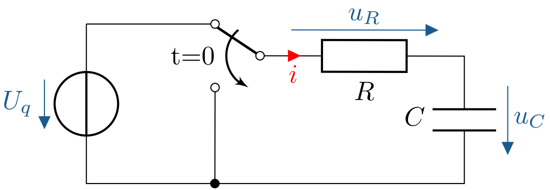




given: $u_C(0)$, U_q, R, C

wanted: $u_C(t)$

1. **ODE** $$0 = RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C + u_C$$
2. **Homogeneous solution** $$u_{C,h} = K \cdot \mathrm{e}^{-t/\tau} \quad \text{with } \tau = RC$$
3. **Particular solution** $$u_{C,p} = 0$$
4. **Non-homog. solution** $$u_C = K \cdot \mathrm{e}^{-t/\tau}$$
5. **Determine constants** $$u_C = U_q \cdot \mathrm{e}^{-t/\tau}$$



First-order ODE:

$$a_1 \cdot \frac{\mathrm{d}}{\mathrm{d}t} y(t) + a_0 \cdot y(t) = b$$

1.) Set up ODE: for $u_C(t)$ $(t \geq 0)$

Based on the mesh equation, u_R is expressed as a function of u_C:

Kirchhoff : Mesh: $u_R + u_C = 0$ Knot: $i_R = i_C = i$

$R \cdot i + u_C = 0$ with $u_R = R \cdot i_R = R \cdot i$

ODE : $RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C + u_C = 0$ with $i = i_C = C \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C$

Discharging process In terms of capacity

2.) Homog. solution, gen.: (fleeting)

ODE$_h$: $RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_{C,h} + u_{C,h} = 0$ Approach: $u_{C,h} = K \cdot \mathrm{e}^{\lambda t}$

char. Polynom: $\Rightarrow \quad RC \cdot \lambda + 1 = 0 \quad \Rightarrow \quad \lambda = -\frac{1}{RC}$

hom. Sol. : $u_{C,h} = K \cdot \mathrm{e}^{-t/\tau}$ with $\tau = RC$

3.) Part. Sol., 4.) Superposition: (settled zero)

allg. Lsg.: $u_C = u_{C,h} + \cancel{u_{C,p}}$ with $u_{C,p} = 0$

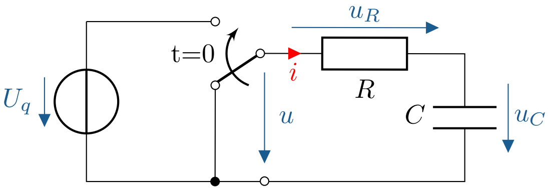
5.) Determine constants: (Insert AB)

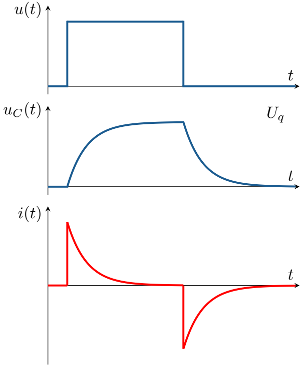
Initial conditions. $u_C(0) = K \cdot \mathrm{e}^0 \stackrel{!}{=} U_q \quad \Rightarrow \quad K = U_q$

special gen. sol.: $u_C(t) = U_q \cdot \mathrm{e}^{-t/\tau}$

Current: $i_C(t) = -\frac{U_q}{R} \cdot \mathrm{e}^{-t/\tau}$ with $i_C = C \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C$

Comparison of loading and unloading processes In terms of capacity



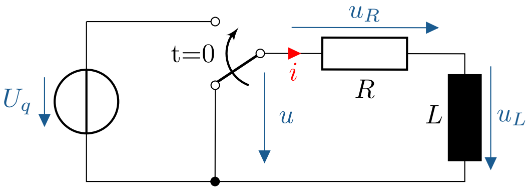


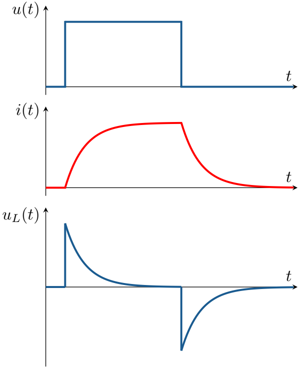
Calculation methods are basically the same for (DC)
charging and discharging processes.

Voltage u_C after switching time t_0 [see script]:

$$u_C(t-t_0) = \underbrace{u_C(t_0)}_{\text{Initial value}} + \underbrace{\left(u_{C,\mathrm{e}} - u_C(t_0)\right)}_{\text{Difference}} \cdot \underbrace{\left(1 - \mathrm{e}^{-\frac{t-t_0}{\tau}}\right)}_{\text{Exp. approximation}}$$

Differences only in the disturbance term of the ODE, initial condition, and final value.



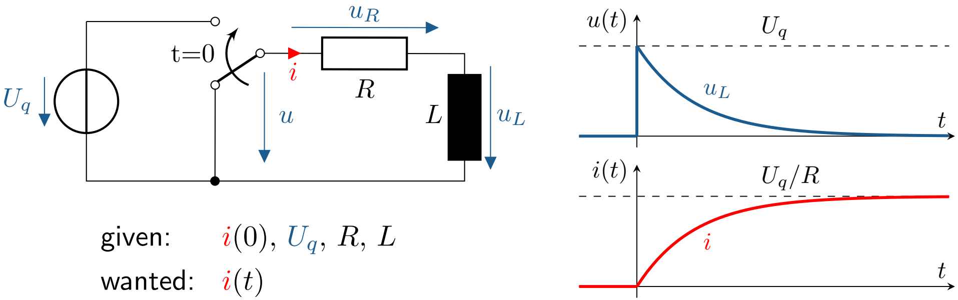


The calculation method for RL elements is basically the same as for RC elements. (First-order ODE) Note: i continuous, u_L discontinuous.

Presumption: Current and voltage „reversed“relative to RC element.

$$i(t-t_0) = \underbrace{i(t_0)}_{\text{Initial value}} + \underbrace{\big(i_\mathrm{e} - i(t_0)\big)}_{\text{Difference}} \cdot \underbrace{\left(1 - \mathrm{e}^{-\frac{t-t_0}{\tau}}\right)}_{\text{Exp. approximation}}$$

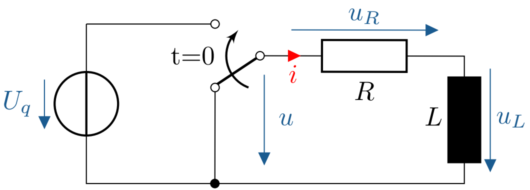
Charging: i increasingly, u_L decreasingly. **Discharging:** vice versa.



given: $i(0)$, U_q, R, L

wanted: $i(t)$

1. **ODE** $$\frac{U_q}{R} = \frac{L}{R} \cdot \frac{\mathrm{d}}{\mathrm{d}t} i + i$$
2. **Homogeneous Sol.** $$i_\mathrm{h} = K \cdot \mathrm{e}^{-t/\tau} \quad \text{with} \quad \tau = \frac{L}{R}$$
3. **Particular Sol.** $$i_\mathrm{p} = \frac{U_q}{R}$$
4. **Non-homogeneous sol.** $$i = K \cdot \mathrm{e}^{-t/\tau} + \frac{U_q}{R}$$
5. **Determine const.** $$i = \frac{U_q}{R}\left(1 - \mathrm{e}^{-t/\tau}\right)$$



First-order ODE:

$$a_1 \cdot \frac{\mathrm{d}}{\mathrm{d}t} y(t) + a_0 \cdot y(t) = b$$

1. Set up ODE: for $i(t)$ $(t \geq 0)$

Mesh : $u_R + u_L = U_q$ with $u_R = R \cdot i$

$$R \cdot i + L \cdot \frac{\mathrm{d}}{\mathrm{d}t} i = U_q \quad \text{with} \quad u_L = L \cdot \frac{\mathrm{d}}{\mathrm{d}t} i$$

$$L \cdot \frac{\mathrm{d}}{\mathrm{d}t} i + R \cdot i = U_q \quad \text{with} \quad u_L = L \cdot \frac{\mathrm{d}}{\mathrm{d}t} i$$

ODE : $$\frac{L}{R} \cdot \frac{\mathrm{d}}{\mathrm{d}t} i + i = \frac{U_q}{R}$$

Charging process For Inductances

2.) Homogenous Solution, gen.: (fleeting) (without interference)

ODE$_h$: $\frac{L}{R} \cdot \frac{\mathrm{d}}{\mathrm{d}t} i_\mathrm{h} + i_\mathrm{h} = 0$ Approach: $i_\mathrm{h} = K \cdot \mathrm{e}^{\lambda t}$

char. Pol. : $\frac{L}{R} \cdot \lambda + 1 = 0$ $\Rightarrow \quad \lambda = -\frac{1}{\tau} = -\frac{R}{L}$

hom. Sol. : $i_\mathrm{h} = K \cdot \mathrm{e}^{-t/\tau}$ with $\tau = \frac{L}{R}$

2.) Homogenous Solution, gen.: (fleeting) (without interference)

$$\text{ODE}_h: \quad \frac{L}{R} \cdot \frac{\mathrm{d}}{\mathrm{d}t} i_\mathrm{h} + i_\mathrm{h} = 0 \qquad \text{Approach:} \quad i_\mathrm{h} = K \cdot \mathrm{e}^{\lambda t}$$

$$\text{char. Pol.}: \quad \frac{L}{R} \cdot \lambda + 1 = 0 \qquad \Rightarrow \quad \lambda = -\frac{1}{\tau} = -\frac{R}{L}$$

$$\text{hom. Sol.}: \quad i_\mathrm{h} = K \cdot \mathrm{e}^{-t/\tau} \qquad \text{with} \quad \tau = \frac{L}{R}$$

3.) Particular solution: (settled in) with $t \to \infty$

$$\text{ODE}_p: \quad \cancel{\frac{L}{R} \cdot \frac{\mathrm{d}}{\mathrm{d}t} i_\mathrm{p}} + i_\mathrm{p} = \frac{U_q}{R} \qquad \text{with} \quad i_\mathrm{p} = const.$$

$$\text{part. Sol.}: \quad i_\mathrm{p} = \frac{U_q}{R}$$

4.) Non-homogeneous sol., gen.: (Superimposition)

gener. Sol.: $$i(t) = \underbrace{K \cdot \mathrm{e}^{-t/\tau}}_{i_\mathrm{h}} + \underbrace{\frac{U_q}{R}}_{i_\mathrm{p}}$$ with $i_\mathrm{h} = i_f, \quad i_\mathrm{p} = i_e$

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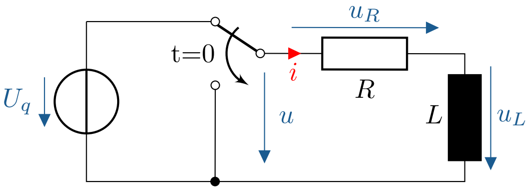
5.) Determine constants: (Set initial conditions)

initial cond. $$i(0) = K \cdot \cancel{\mathrm{e}^0} + \frac{U_q}{R} \stackrel{!}{=} 0 \qquad \Rightarrow \qquad K = -\frac{U_q}{R}$$

clear solution: $$i(t) = \frac{U_q}{R} \cdot \left(1 - \mathrm{e}^{-t/\tau}\right)$$

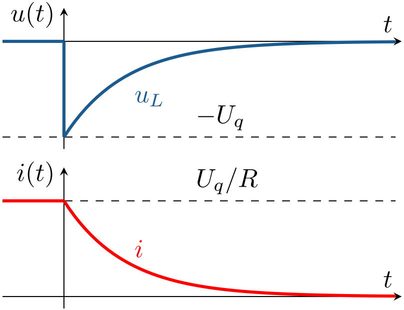
Determine voltage u_L from current i:

Voltage: $$u_L = U_q \cdot \mathrm{e}^{-t/\tau} \qquad \text{with} \quad u_L = L \cdot \frac{\mathrm{d}}{\mathrm{d}t} i$$



given: $i(0)$, U_q, R, L

wanted: $i(t)$



1. ODE $$0 = \frac{L}{R} \cdot \frac{\mathrm{d}}{\mathrm{d}t} i + i$$

2. Homogenous sol. $$i_\mathrm{h} = K \cdot \mathrm{e}^{-t/\tau} \quad \text{with} \quad \tau = \frac{L}{R}$$

3. Particular sol. $$i_\mathrm{p} = 0$$

4. Non-homogeneous sol. $$i = K \cdot \mathrm{e}^{-t/\tau}$$

5. Determine constants $$i = \frac{U_q}{R} \cdot \mathrm{e}^{-t/\tau}$$

Switching operations With sinusoidal excitation

Switching operations with AC excitation are:

- frequency-dependent (f, ω)
- switching time dependent (t_0)

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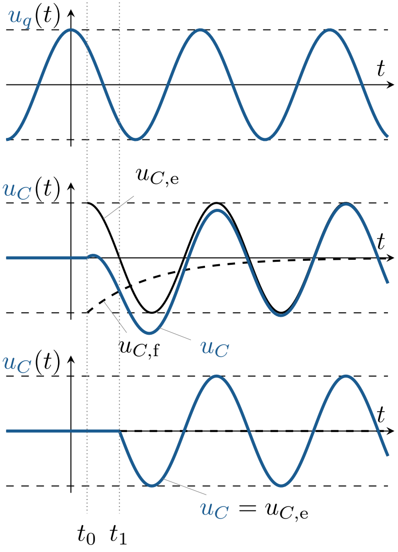
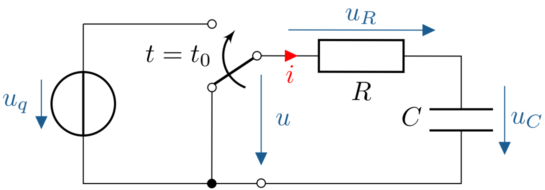


Comparison: switching times t_0 and t_1

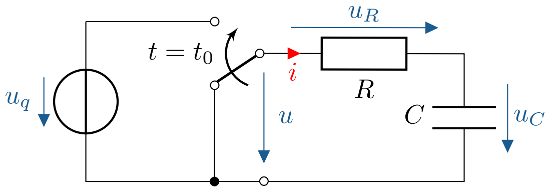
Switching operations With sinusoidal excitation In terms of capacity

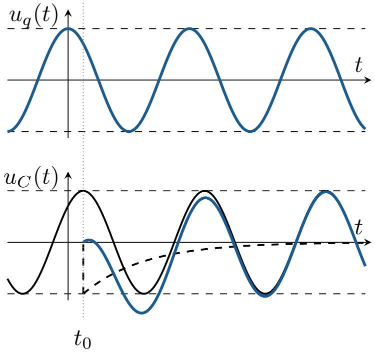


given: $u_C(t_0)$, $u_q(t)$, R, C, ω

wanted: $u_C(t)$ for $t \geq t_0$

Switching operations With sinusoidal excitation In terms of capacity





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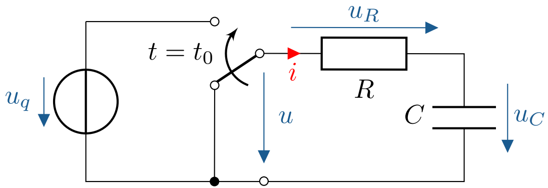
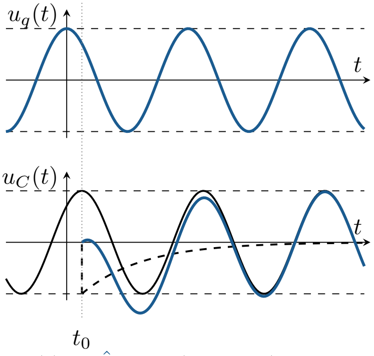
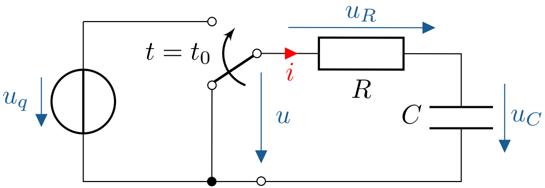




given: $u_C(t_0)$, $u_q(t)$, R, C, ω

wanted: $u_C(t)$ for $t \geq t_0$

1. **ODE** $$u_q = RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C(t) + u_C(t) = \hat{U}_q \cdot \cos(\omega t + \varphi_q)$$
2. **Homogeneous solution** $$u_{C,\mathrm{h}} = K \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}} \quad \text{with} \quad \tau = RC$$
3. **Particular solution** $$u_{C,\mathrm{p}} = \hat{U}_{C,\mathrm{p}} \cdot \cos(\omega t + \varphi_{C,\mathrm{p}})$$
4. **Inhomogeneous solutiong** $$u_C = K \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}} + \hat{U}_{C,\mathrm{p}} \cdot \cos(\omega t + \varphi_{C,\mathrm{p}})$$
5. **Determine constants** $$u_C = \hat{U}_{C,\mathrm{p}} \cdot \left(-\cos(\omega t_0 + \varphi_{C,\mathrm{p}}) \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}} + \cos(\omega t + \varphi_{C,\mathrm{p}})\right)$$



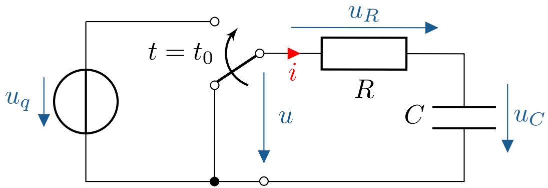
First-order ODE:

$$a_1 \cdot \frac{\mathrm{d}}{\mathrm{d}t} y(t) + a_0 \cdot y(t) = b$$

1.) Set up ODE: for u_c $(t \geq t_0)$

$$RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C(t) + u_C(t) = u_q(t)$$

with $\quad u_q(t) = \hat{U}_q \cdot \cos(\omega t + \varphi_q) \quad$ (Disturbance)



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with $\quad u_q(t) = \hat{U}_q \cdot \cos(\omega t + \varphi_q) \quad$ (Disturbance)

2.) Homogeneous solution, general (fleeting) (without interference):

$$u_{C,\mathrm{h}} = K \cdot \mathrm{e}^{-\frac{t'}{\tau}} \qquad \text{with} \quad t' = t - t_0$$

$$= K \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}} \qquad \text{with} \quad \tau = RC$$

3.) Particular solution (settled in) ($t \to \infty$): sinusoidal!

Application of complex alternating current calculation:

$$u_{C,p} = \hat{U}_{C,p} \cdot \cos(\omega t + \varphi_{C,\mathrm{p}})$$
$$= \mathrm{Re}\left\{\underline{\hat{U}}_{C,p} \cdot \mathrm{e}^{\mathrm{j}\omega t}\right\} \qquad \text{with} \quad \underline{\hat{U}}_{C,p} = \hat{U}_{C,p} \cdot \mathrm{e}^{\mathrm{j}\varphi_{C,\mathrm{p}}}$$

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Complex amplitude indicator $\underline{\hat{U}}_{C,p}$ using complex voltage divider:

$$\underline{\hat{U}}_{C,p} = \underline{\hat{U}}_q \cdot \frac{\underline{Z}_C}{R + \underline{Z}_C} = \underline{\hat{U}}_q \cdot \frac{\frac{1}{\mathrm{j}\omega C}}{R + \frac{1}{\mathrm{j}\omega C}} = \frac{\underline{\hat{U}}_q}{1 + \mathrm{j}\omega C R} \qquad \text{with} \quad \underline{\hat{U}}_q = \hat{U}_q \cdot \mathrm{e}^{\mathrm{j}\varphi_q}$$

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For the amplitude $\hat{U}_{C,p}$ and the (load) phase angle $\varphi_{C,\mathrm{p}}$, the following applies:

$$\hat{U}_{C,p} = \frac{\hat{U}_q}{\sqrt{1 + (\omega CR)^2}} \qquad \text{und} \qquad \varphi_{C,\mathrm{p}} = \varphi_q - \arctan(\omega CR)$$

4.) Inhomogeneous solution, general (Superposition):

$$\begin{aligned} u_C &= u_{C,\mathrm{h}} + u_{C,p} \\ &= K \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}} + \hat{U}_{C,p} \cdot \cos(\omega t + \varphi_{C,\mathrm{p}}) \end{aligned}$$

5.) Determine constants (Initial condition):

$$\begin{aligned} u_C(t_0) &\overset{!}{=} 0 \\ K \cdot \cancel{\mathrm{e}^0} + \hat{U}_{C,p} \cdot \cos(\omega t_0 + \varphi_{C,\mathrm{p}}) &= 0 \\ K &= -\hat{U}_{C,p} \cdot \cos(\omega t_0 + \varphi_{C,\mathrm{p}}) \end{aligned}$$

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\begin{aligned}
u_C &= u_{C,\mathrm{h}} + u_{C,p} \\
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\end{aligned}
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$$
\begin{aligned}
u_C(t_0) &\overset{!}{=} 0 \\
K \cdot \cancel{\mathrm{e}^0} + \hat{U}_{C,p} \cdot \cos(\omega t_0 + \varphi_{C,\mathrm{p}}) &= 0 \\
K &= -\hat{U}_{C,p} \cdot \cos(\omega t_0 + \varphi_{C,\mathrm{p}})
\end{aligned}
$$

The unique solution is obtained by substituting K into the general solution:

$$
u_C(t) = \underbrace{-\hat{U}_{C,p} \cdot \cos(\omega t_0 + \varphi_{C,\mathrm{p}}) \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}}}_{u_{C,\mathrm{h}}} + \underbrace{\hat{U}_{C,p} \cdot \cos\left(\omega t + \varphi_{C,\mathrm{p}}\right)}_{u_{C,p}}
$$

For sinusoidal excitation:

With large time constants, **overvoltages/overcurrents are possible**.

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Excess values of up to twice/three times the peak
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Switching operations With sinusoidal excitation In terms of capacity

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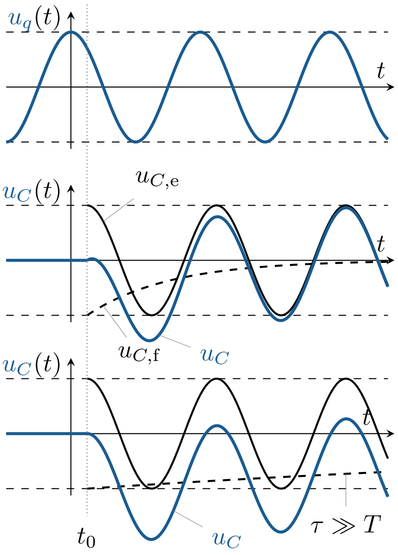
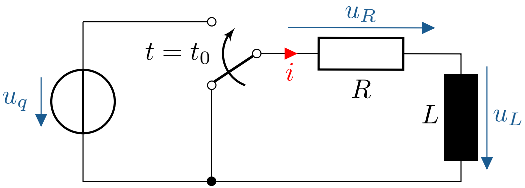
Example: Disconnect switch in the network



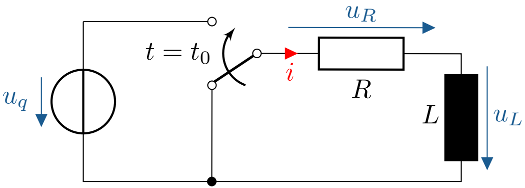
Comparison: small and large τ



given: $i(t_0)$, $u_q(t)$, R, L, ω

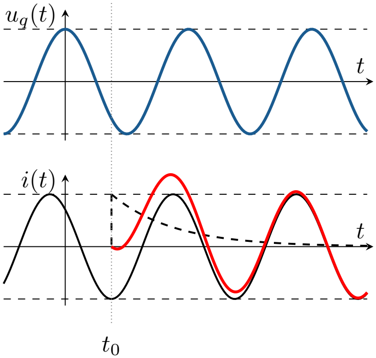
wanted: $i(t)$ for $t \geq t_0$

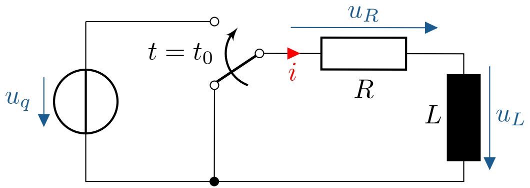
Switching operations With sinusoidal excitation For Inductors

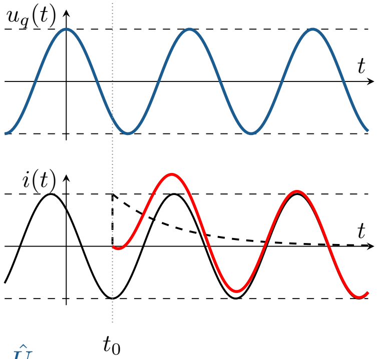


given: $i(t_0)$, $u_q(t)$, R, L, ω

wanted: $i(t)$ for $t \geq t_0$







given: $i(t_0)$, $u_q(t)$, R, L, ω

wanted: $i(t)$ for $t \geq t_0$

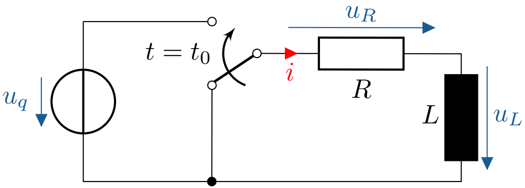
1. ODE $$\frac{u_q}{R} = \frac{L}{R} \cdot \frac{\mathrm{d}}{\mathrm{d}t} i + i = \frac{\hat{U}_q}{R} \cdot \cos(\omega t + \varphi_q)$$

2. Homogeneous sol. $$i_\mathrm{h} = K \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}} \quad \text{with} \quad \tau = \frac{L}{R}$$

3. Particular sol. $$i_\mathrm{p} = \hat{I}_\mathrm{p} \cdot \cos(\omega t + \varphi_I)$$

4. Non-homogeneous sol. $$i = K \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}} + \hat{I}_\mathrm{p} \cdot \cos(\omega t + \varphi_I)$$

5. Determine constants $$i = \hat{I}_\mathrm{p} \cdot \left(- \cos(\omega t_0 + \varphi_I) \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}} + \cos(\omega t + \varphi_I) \right)$$

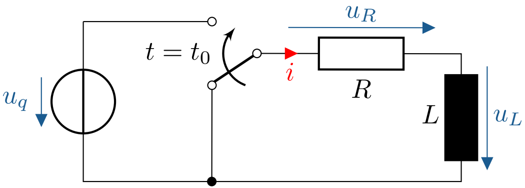


ODE 1st order:

$$a_1 \cdot \frac{\mathrm{d}}{\mathrm{d}t} y(t) + a_0 \cdot y(t) = b$$

1.) Set up ODE: for i with $t \geq t_0$

ODE : $$\frac{L}{R} \cdot \frac{\mathrm{d}}{\mathrm{d}t} i + i = \frac{u_q}{R}$$ with $u_q = \hat{U}_q \cdot \cos(\omega t + \varphi_q)$



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2.) Homogeneous solution, gen.: (fleeting, for switching time t_0)

hom. Sol.: $$i_\mathrm{h} = K \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}}$$ with $\tau = \frac{L}{R}$

3.) Particular solution: (settled in, with $t \to \infty$)

ODE transf. : $$\frac{L}{R} \cdot \mathrm{j}\omega\, \underline{\hat{I}}_{\mathrm{p}} + \underline{\hat{I}}_{\mathrm{p}} = \frac{\underline{\hat{U}}_q}{R}$$ with $\underline{\hat{U}}_q = \hat{U}_q \cdot \mathrm{e}^{\mathrm{j}\varphi_q}$

Pointer: $$\underline{\hat{I}}_{\mathrm{p}} = \frac{\underline{\hat{U}}_q}{R} \cdot \frac{1}{1 + \mathrm{j}\omega \cdot \frac{L}{R}}$$ with $\underline{\hat{I}}_{\mathrm{p}} = \hat{I}_{\mathrm{p}} \cdot \mathrm{e}^{\mathrm{j}\varphi_I}$

$$\hat{I}_{\mathrm{p}} = \frac{\hat{U}_q}{R} \cdot \frac{1}{\sqrt{1 + \left(\omega \cdot \frac{L}{R}\right)^2}}$$

$$\varphi_I = \varphi_q - \arctan\left(\omega \cdot \frac{L}{R}\right)$$

part. Sol. : $$i_{\mathrm{p}} = \hat{I}_{\mathrm{p}} \cdot \cos(\omega t + \varphi_I)$$ with $i_{\mathrm{p}} = \mathrm{Re}\left\{\underline{\hat{I}}_{\mathrm{p}} \cdot \mathrm{e}^{\mathrm{j}\omega t}\right\}$

4.) Non-homogeneous solution, gen.: (Superposition)

Gen. Sol.:

$$i = \underbrace{K \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}}}_{i_\mathrm{h}} + \underbrace{\hat{I}_\mathrm{p} \cdot \cos(\omega t + \varphi_I)}_{i_\mathrm{p}}$$

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5.) Determine constants: (Insert AB)

Initial requi.:
$$i(t_0) = K \cdot \cancel{\mathrm{e}^0} + \hat{I}_\mathrm{p} \cdot \cos(\omega t_0 + \varphi_I) \stackrel{!}{=} 0$$
$$\Rightarrow \quad K = -\hat{I}_\mathrm{p} \cdot \cos(\omega t_0 + \varphi_I)$$

single sol.:
$$i(t) = -\hat{I}_\mathrm{p} \cdot \cos(\omega t_0 + \varphi_I) \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}} + \hat{I}_\mathrm{p} \cdot \cos(\omega t + \varphi_I)$$

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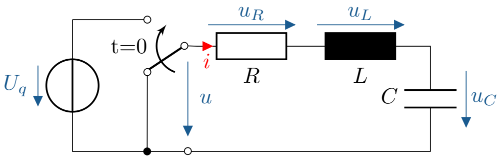
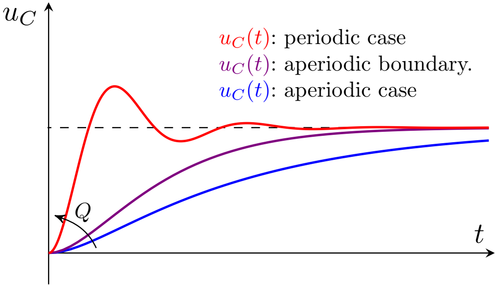
Voltage u_L with $u_L = L \cdot \frac{\mathrm{d}}{\mathrm{d}t} i$ calculate:

Voltage: $$u_L(t) = R \cdot \hat{I}_\mathrm{p} \cdot \cos(\omega t_0 + \varphi_I) \cdot \mathrm{e}^{-\frac{t-t_0}{\tau}} - \omega L \cdot \hat{I}_\mathrm{p} \cdot \sin(\omega t + \varphi_I)$$



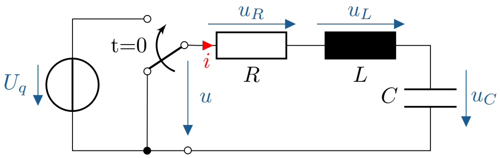


Because of L and C $\Rightarrow$ oscillating circuit

ODE (for u_C for $t \geq 0$):

$$LC \cdot \frac{\mathrm{d}^2}{\mathrm{d}t^2} u_C + RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C + u_C = U_q$$

Case differentiation	Damping	Discri.	Güte
Aperiodic case	strong	$D > 0$	$Q < 0.5$
Aperiodic borderline case	critical	$D = 0$	$Q = 0.5$
Periodic case	weak	$D < 0$	$Q > 0{,}5$



Approach: $u_{C,\mathrm{h}} = K \cdot \mathrm{e}^{\lambda t}$

1.) ODE:

$$u_L + u_R + u_C = u$$

$$L \cdot \frac{\mathrm{d}}{\mathrm{d}t} i + R \cdot i + u_C = U_q$$

$$LC \cdot \frac{\mathrm{d}^2}{\mathrm{d}t^2} u_C + RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_C + u_C = U_q$$

homogeneous ODE:

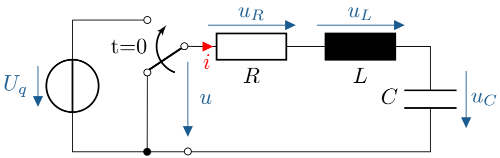
$$LC \cdot \frac{\mathrm{d}^2}{\mathrm{d}t^2} u_{C,\mathrm{h}} + RC \cdot \frac{\mathrm{d}}{\mathrm{d}t} u_{C,\mathrm{h}} + u_{C,\mathrm{h}} = 0$$

char. Polynom:

$$\lambda^2 + \frac{R}{L} \cdot \lambda + \frac{1}{LC} = 0$$

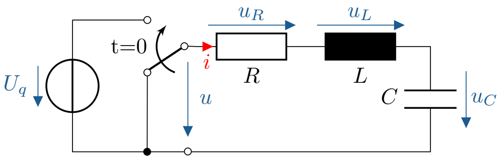
Eigenvalue:

$$\lambda_{1/2} = -\frac{R}{2L} \pm \sqrt{\underbrace{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}_{\text{Discriminant } D}}$$



$$\lambda_{1/2} = -\frac{R}{2L} \pm \sqrt{\underbrace{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}_{\text{Discriminant } D}}$$

Attention! Depending on the discriminant D, different eigenvalues $\lambda_{1/2}$.



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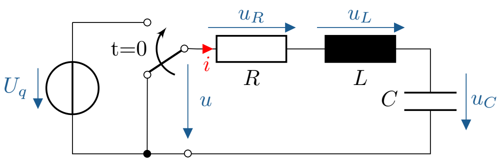
2.) Homogeneous solution:

$$u_{C,\mathrm{h}} = \begin{cases} K_1 \cdot \mathrm{e}^{\lambda_1 t} + K_2 \cdot \mathrm{e}^{\lambda_2 t} & \text{for} \quad \lambda_1 \neq \lambda_2 \\ (K_1 + K_2 \cdot t) \cdot \mathrm{e}^{\lambda t} & \text{for} \quad \lambda_1 = \lambda_2 = \lambda \end{cases}$$

Case differentiation for RLC series resonant circuits



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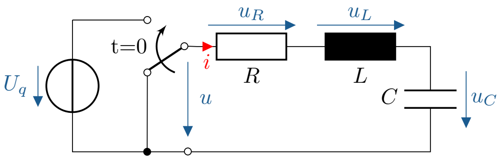
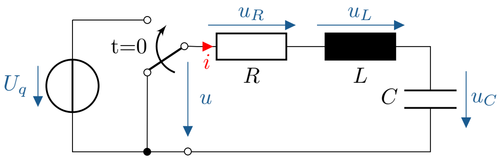
Case differentiation	**Discrimi.**	**Eigenvalue $\lambda_{1/2}$**
aperiodic case	$D > 0$	Two different, real NSTs
aperiodic borderline case	$D = 0$	A double, real NST
periodic case	$D < 0$	Two conjugated, complex NSTs



$$\lambda_{1/2} = -\frac{R}{2L} \pm \underbrace{\sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}}_{\text{Discriminant } D}$$

Aperiodic case (Creep case)

with $D > 0$ applies $\quad \frac{R}{2L} \overset{!}{>} \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \quad \Rightarrow$ 2 real eigenvalues



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2.) Homogeneous solution: (fleeting)

$$u_{C,\mathrm{h}} = K_1 \cdot \mathrm{e}^{\lambda_1 t} + K_2 \cdot \mathrm{e}^{\lambda_2 t} \qquad \text{with} \quad \lambda_1 \neq \lambda_2,\ \lambda < 0 \tag{1}$$

3.) Particular solution: (settled, identical for all three cases)

$$u_{C,\mathrm{p}} = U_q$$

Aperiodic case in the RLC series resonant circuit

4.) General solution: (Superposition)

$$u_C(t) = K_1 \cdot \mathrm{e}^{\lambda_1 t} + K_2 \cdot \mathrm{e}^{\lambda_2 t} + U_q$$

4.) General solution: (Superposition)

$$u_C(t) = K_1 \cdot \mathrm{e}^{\lambda_1 t} + K_2 \cdot \mathrm{e}^{\lambda_2 t} + U_q$$

5.) Determine constants: (as is known)

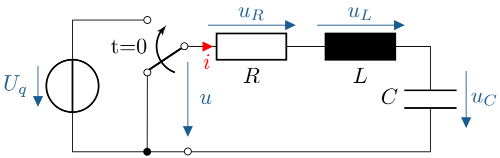
Initial conditions (AB): $u_C(0) = 0$ and $i(0) = 0 \Rightarrow \frac{\mathrm{d}}{\mathrm{d}t} u_C(0) = 0$

1st IC: $$u_C(0) = K_1 \cdot \cancel{\mathrm{e}^{\lambda_1 0}} + K_2 \cdot \cancel{\mathrm{e}^{\lambda_2 0}} + U_q \overset{!}{=} 0 \quad \Rightarrow \quad K_1 = -K_2 - U_q$$

2nd IC: $$\frac{\mathrm{d}}{\mathrm{d}t} u_C(0) = K_1 \cdot \lambda_1 \cdot \cancel{\mathrm{e}^{\lambda_1 0}} + K_2 \cdot \lambda_2 \cdot \cancel{\mathrm{e}^{\lambda_2 0}} + \cancel{0} \overset{!}{=} 0 \quad \Rightarrow \quad K_1 = -K_2 \frac{\lambda_2}{\lambda_1}$$

$$\Longrightarrow \qquad K_1 = \frac{U_q}{\frac{\lambda_1}{\lambda_2} - 1} \quad \text{and} \quad K_2 = \frac{U_q}{\frac{\lambda_2}{\lambda_1} - 1}$$

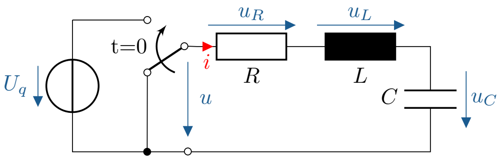
Aperiodic boundary case in the RLC series resonant circuit



$$\lambda_{1/2} = -\frac{R}{2L} \pm \sqrt{\underbrace{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}_{\text{Discriminant } D}}$$

Aperiodic borderline case:

for $D = 0$ applies $\lambda_{1/2} = -\frac{R}{2L} \cancel{\pm \sqrt{0}} = \lambda$ $\Rightarrow$ 1 real Eigenvalue



$$\lambda_{1/2} = -\frac{R}{2L} \pm \sqrt{\underbrace{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}_{\text{Discriminant } D}}$$

Aperiodic borderline case:

for $D = 0$ applies $\lambda_{1/2} = -\frac{R}{2L} \cancel{\pm \sqrt{0}} = \lambda$ $\Rightarrow$ 1 real Eigenvalue

2.) Homogeneous solution: (fleeting)

$$u_{C,\mathrm{h}} = (K_1 + K_2 \cdot t) \cdot \mathrm{e}^{\lambda t} \qquad \text{with} \quad \lambda_1 = \lambda_2 < 0 \tag{2}$$

3.) Particular solution: (settled in)

$$u_{C,\mathrm{p}} = U_q$$

4.) General solution: (settled in)

$$u_C(t) = (K_1 + K_2 \cdot t) \cdot \mathrm{e}^{\lambda t} + U_q$$

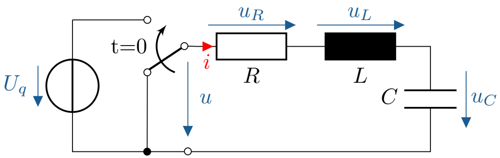
5.) Determine constants: (as is known)

Initial conditions (AB): $u_C(0) = 0$ and $i(0) = 0 \Rightarrow \frac{\mathrm{d}}{\mathrm{d}t} u_C(0) = 0$

1st IC: $u_C(0) = (K_1 + \cancel{K_2 \cdot 0}) \cancel{\cdot \mathrm{e}^{\lambda 0}} + U_q \stackrel{!}{=} 0 \quad \Rightarrow \quad K_1 = -U_q$

2nd IC: $\dfrac{\mathrm{d}}{\mathrm{d}t} u_C(0) = (K_1 + \cancel{K_2 \cdot 0}) \cdot \lambda \cdot \cancel{\mathrm{e}^{\lambda 0}} + K_2 \cdot \cancel{\mathrm{e}^{\lambda 0}} \stackrel{!}{=} 0 \quad \Rightarrow \quad K_2 = -K_1 \cdot \lambda$

$\Longrightarrow \quad K_1 = -U_q$ and $K_2 = +U_q \cdot \lambda$

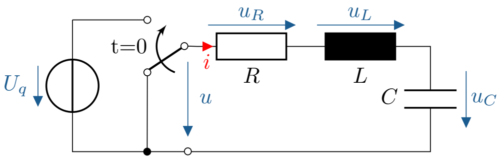


$$\lambda_{1/2} = -\frac{R}{2L} \pm \underbrace{\sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}}_{\text{Discriminant } D}$$

Periodic case: (swing fall)

for $D < 0$ applies $$\lambda_{1/2} = -\underbrace{\frac{R}{2L}}_{\delta} \pm \mathrm{j} \cdot \underbrace{\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}}_{\omega_d} \quad \Rightarrow \quad \text{2 compl. EW} \tag{3}$$

Periodic case for the RLC series resonant circuit



$$\lambda_{1/2} = -\frac{R}{2L} \pm \sqrt{\underbrace{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}_{\text{Discriminant } D}}$$

Periodic case: (swing fall)

$$\text{for} \quad D < 0 \quad \text{applies} \qquad \lambda_{1/2} = -\underbrace{\frac{R}{2L}}_{\delta} \pm \mathrm{j} \cdot \underbrace{\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}}_{\omega_d} \qquad \Rightarrow \qquad \text{2 compl. EW} \tag{3}$$

2.) Homogeneous solution: (fleeting)

$$u_{C,\mathrm{h}} = \mathrm{e}^{-\delta t} \cdot C_0 \cdot \sin(\omega_d t + \varphi_0) \qquad \text{with} \quad \lambda_{1/2} = -\delta \pm \mathrm{j}\,\omega_d \tag{4}$$

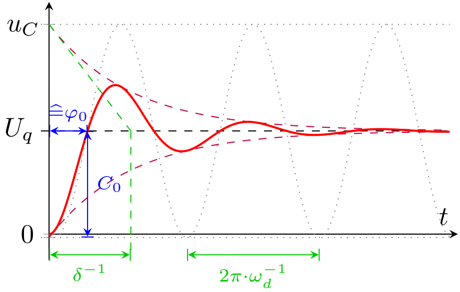
Derivation for $u_{C,\mathrm{h}}$ in Excursus. Determination of the constants C_0 and φ_0 as known.

3.) Particular solution: (settled in)

$$u_{C,\mathrm{p}} = U_q$$

4.) General solution: (Superposition)

$$u_C(t) = \mathrm{e}^{-\delta t} \cdot C_0 \cdot \sin(\omega_d t + \varphi_0) + U_q$$



5.) Determine constants: (as known)

1. AB: $$u_C(0) = \cancel{\mathrm{e}^{-\delta 0}} \cdot C_0 \cdot \sin(\cancel{\omega_d t} + \varphi_0) + U_q \overset{!}{=} 0$$

2. AB: $$\frac{\mathrm{d}}{\mathrm{d}t} u_C(0) = -\delta \cdot \cancel{\mathrm{e}^{-\delta 0}} \cdot C_0 \cdot \sin(\varphi_0) + \cancel{\mathrm{e}^{-\delta 0}} \cdot C_0 \cdot \omega_d \cdot \cos(\varphi_0) \overset{!}{=} 0$$

$$\Longrightarrow \qquad C_0 = -\frac{U_q}{\sin(\varphi_0)} \quad \text{and} \quad \varphi_0 = \arctan\left(\frac{\omega_d}{\delta}\right)$$

Excursion: Derivation of the homogeneous solution in the periodic case

$$u_{C,\mathrm{h}} = \mathrm{e}^{-\delta t} \cdot \left(K_1 \cdot \mathrm{e}^{+\mathrm{j}\omega_d t} + K_2 \cdot \mathrm{e}^{-\mathrm{j}\omega_d t}\right) \tag{5}$$

K_1 and K_2 must be complex conjugates so that $u_{C,\mathrm{h}}$ is real!

with $\quad K_1 = a + \mathrm{j}b \quad$ and $\quad K_2 = a - \mathrm{j}b \quad$ with $\quad a,b \in \mathbb{R}$

$$u_{C,\mathrm{h}} = \mathrm{e}^{-\delta t} \cdot \left[(a + \mathrm{j}b) \cdot \mathrm{e}^{+\mathrm{j}\omega_d t} + (a - \mathrm{j}b) \cdot \mathrm{e}^{-\mathrm{j}\omega_d t}\right]$$

$$u_{C,\mathrm{h}} = \mathrm{e}^{-\delta t} \cdot \left[a \cdot \underbrace{\left(\mathrm{e}^{+\mathrm{j}\omega_d t} + \mathrm{e}^{-\mathrm{j}\omega_d t}\right)}_{2\cos(\omega_d t)} + \mathrm{j}b \cdot \underbrace{\left(\mathrm{e}^{+\mathrm{j}\omega_d t} - \mathrm{e}^{-\mathrm{j}\omega_d t}\right)}_{2\mathrm{j}\sin(\omega_d t)}\right]$$

with

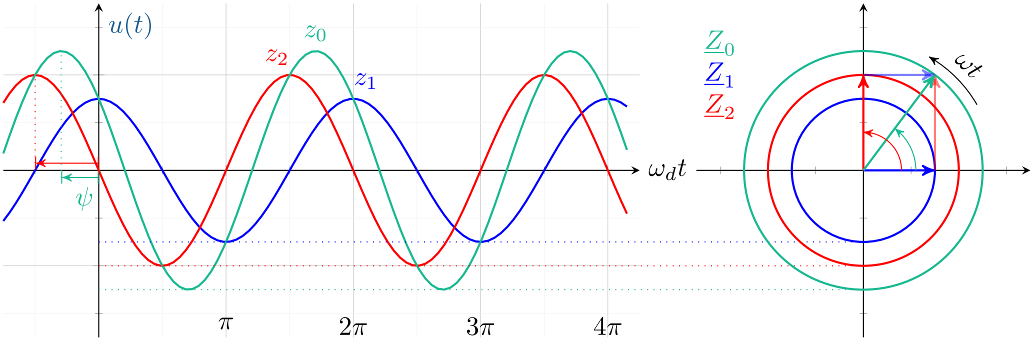
$$\mathrm{e}^{+\mathrm{j}x} + \mathrm{e}^{-\mathrm{j}x} = \cos(x) + \cancel{\mathrm{j}\sin(x)} + \cos(x) - \cancel{\mathrm{j}\sin(x)} = 2\cos(x)$$

$$\mathrm{e}^{+\mathrm{j}x} - \mathrm{e}^{-\mathrm{j}x} = \cancel{\cos(x)} + \mathrm{j}\sin(x) - \cancel{\cos(x)} + \mathrm{j}\sin(x) = 2\mathrm{j}\sin(x)$$

$$u_{C,\mathrm{h}} = \mathrm{e}^{-\delta t} \cdot \left[\underbrace{2a}_{C_1} \cdot \cos(\omega_d t) - \underbrace{2b}_{C_2} \cdot \sin(\omega_d t)\right] \tag{6}$$

Superposition of two oscillations with the same angular frequency ω_d:

$$u_{C,\mathrm{h}} = \mathrm{e}^{-\delta t} \cdot \Big[\underbrace{2a}_{C_1} \cdot \cos(\omega_d t) - \underbrace{2b}_{C_2} \cdot \sin(\omega_d t)\Big]$$



$$u_{C,\mathrm{h}} = \mathrm{e}^{-\delta t} \cdot C_0 \cdot \sin(\omega_d t + \varphi_0) \qquad \text{with} \quad \lambda_{1/2} = -\delta \pm \mathrm{j} \cdot \omega_d$$

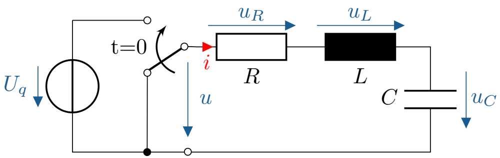
$$u_{C,\mathrm{h}} = \mathrm{e}^{-\delta t} \cdot \Big[\underbrace{2a}_{C_1} \cdot \cos(\omega_d t) - \underbrace{2b}_{C_2} \cdot \sin(\omega_d t)\Big]$$

Time domain		*Polarform*	*Cartesian*
$z_1(t) = +C_1 \cdot \cos(\omega_d t)$	$\Leftrightarrow$	$\underline{z}_1 = C_1 \cdot \mathrm{e}^{\mathrm{j}0}$	$\underline{z}_1 = C_1$
$z_2(t) = -C_2 \cdot \sin(\omega_d t)$	$\Leftrightarrow$	$\underline{z}_2 = C_2 \cdot \mathrm{e}^{-\mathrm{j}\frac{\pi}{2}}$	$\underline{z}_2 = \quad - \mathrm{j} \cdot C_2$
$z_0(t) = z_1(t) + z_2(t)$	$\Leftrightarrow$	$\underline{z}_0 = \underline{z}_1 + \underline{z}_2$	$\underline{z}_0 = C_1 - \mathrm{j} \cdot C_2$
$= C_0 \cdot \cos(\omega_d t + \psi)$	$\Leftrightarrow$	$= C_0 \cdot \mathrm{e}^{\mathrm{j}\psi}$	

with $\quad C_0 = \sqrt{C_1^2 + C_2^2}$ $\quad$ and $\quad \psi = \arctan\left(-\dfrac{C_2}{C_1}\right), \quad \varphi_0 = \psi + \dfrac{\pi}{2}$

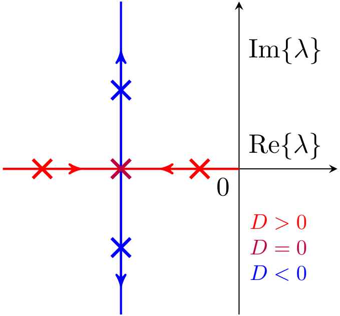
$$z_0(t) = C_0 \cdot \sin(\omega_d t + \varphi_0) \tag{7}$$

$$u_{C,\mathrm{h}} = \mathrm{e}^{-\delta t} \cdot C_0 \cdot \sin(\omega_d t + \varphi_0) \qquad \text{with} \quad \lambda_{1/2} = -\delta \pm \mathrm{j} \cdot \omega_d$$



$$\lambda_{1/2} = -\frac{R}{2L} \pm \sqrt{\underbrace{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}_{\text{Discriminant } D}}$$

Zero points in the locus diagram: $\lambda(D)$ for $\delta = \frac{R}{2L} = const.$



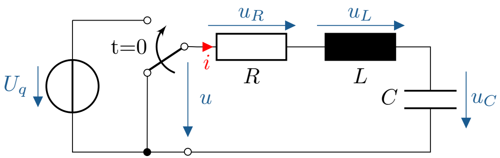
$\mathrm{Re}\{\lambda\} < 0$: $\Rightarrow$ stable

$D > 0$: NST closer to the Im-axis
$\Rightarrow$ slower, dominant

$D = 0$: NST further away from the Im-axis
$\Rightarrow$ fastest aperiodic case

$D < 0$: NST conjugated, complex
$\Rightarrow$ oscillating

Digression: Decay constant, natural and resonance circuit frequency, quality factor



$$\lambda_{1/2} = -\underbrace{\frac{R}{2L}}_{\delta} \pm \mathrm{j} \cdot \underbrace{\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}}_{\omega_d}$$

Periodic case: Relationship between δ, ω_d, ω_0 and Q_S:

Resonant circuit frequency:	$\omega_0 = \sqrt{\frac{1}{LC}}$	with $\mathrm{Im}\{\underline{Z}(\omega_0)\} \overset{!}{=} 0$
Damped natural frequency:	$\omega_d = \sqrt{\omega_0^2 - \delta^2}$	Undamped equal ω_0
Quality factor:	$Q_\mathrm{S} = \frac{X_{L,0}}{R} = \frac{\omega_0 L}{R} = \frac{\omega_0}{2\delta}$	Measure of vibration capability
Comparison of homo. ODEs:	$0 = \ddot{y} + 2\delta \cdot \dot{y} + \omega_0^2 \cdot y$	linearly damped oscillation